

Coronary microvascular dysfunction and cardiovascular outcomes (Multicenter FLOW-CMD Registry): a prospective, multicentre cohort study in South Korea



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Summary

Background Coronary microvascular dysfunction often coexists with epicardial coronary artery disease. Data regarding its prevalence and prognosis in patients undergoing invasive coronary angiography are scarce. This study aimed to evaluate the prevalence and prognosis of coronary microvascular dysfunction in patients undergoing clinically indicated invasive coronary angiography in routine practice.

Methods In this prospective, multicentre cohort study done in seven tertiary medical hospitals in South Korea, consecutive patients aged 18 years and older who were referred for clinically indicated invasive coronary angiography were systematically screened and evaluated by coronary physiological assessment. Obstructive epicardial coronary artery disease was defined as an intermediate stenosis (40–90% diameter stenosis), with fractional flow reserve of 0·80 or less or severe stenosis (>90% of diameter stenosis) treated with revascularisation without fractional flow reserve measurement. Coronary microvascular dysfunction was identified as coronary flow reserve below 2·0 and index of microcirculatory resistance of ≥ 25 . The primary endpoint was a composite of all-cause death, myocardial infarction, clinically driven repeat revascularisation, or hospitalisation for heart failure. The Multicenter FLOW-CMD Registry study is registered with ClinicalTrials.gov (NCT05369182).

Findings Between April 22, 2022, and Nov 19, 2024, 5764 patients were screened and 1003 patients were enrolled (756 men and 247 women). Among these patients, coronary microvascular dysfunction was observed in 123 (21·5%) of 573 patients with obstructive epicardial coronary artery disease and in 40 (9·3%) of 430 patients without obstructive epicardial coronary artery disease. At a median follow-up of 1·9 years, the primary endpoint occurred in 26 patients (2-year Kaplan–Meier estimate 18·8%) with coronary microvascular dysfunction and 70 patients (2-year Kaplan–Meier estimate 10·5%) with preserved microvascular function (hazard ratio 1·91 [95% CI 1·22–2·99]; $p=0·0047$).

Interpretation In patients with suspected ischaemic heart disease undergoing invasive coronary angiography, coronary microvascular dysfunction coexisted with epicardial coronary artery disease and was associated with a higher risk of the composite of all-cause death, myocardial infarction, clinically driven repeat revascularisation, or hospitalisation for heart failure.

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Introduction

The diagnostic and therapeutic strategies for patients with ischaemic heart disease have primarily focused on identifying and treating epicardial coronary artery disease. Current guidelines recommend revascularisation for angiographically severe epicardial coronary stenosis or intermediate stenosis with a fractional flow reserve (FFR) of 0·80 or less.^{1,2} However, coronary circulation comprises three components—conductive epicardial coronary arteries, arterioles, and capillaries^{3,4}—and coronary microvascular dysfunction is an important cause of ischaemic heart disease beyond epicardial coronary artery disease.

Accordingly, 2023 American College of Cardiology and American Heart Association⁵ and 2024 European Society of Cardiology⁶ guidelines suggest that coronary microvascular function can be evaluated with coronary flow reserve (CFR) or the index of microcirculatory resistance (IMR) as a complementary investigation in selected patients. Although previous studies have reported higher rates of adverse clinical events in patients with coronary microvascular dysfunction as defined by CFR or IMR, these findings are not considered definitive due to retrospective study designs,⁷ small sample sizes,^{8,9} and inclusion of highly selected patient populations.^{9–11} We therefore sought to evaluate the prevalence and

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Research in context

Evidence before this study

Clinical evaluation of ischaemic heart disease has traditionally focused on the detection and treatment of epicardial coronary artery disease. However, coronary microvascular dysfunction is recognised as a clinically relevant contributor to ischaemia and adverse cardiovascular outcomes. Contemporary American and European guidelines recommend invasive assessment of coronary microvascular function as a complementary investigation in patients with non-obstructive epicardial coronary artery disease, reflecting the recognition that epicardial disease alone does not fully account for symptoms or risk. We searched PubMed, Embase, and the Cochrane Library for studies published from database inception to Feb 14, 2026, using terms related to “coronary microvascular dysfunction”, “coronary flow reserve”, “index of microcirculatory resistance”, “ischaemic heart disease”, and “clinical outcomes”. We included observational studies and clinical trials evaluating the prognostic implications of coronary microvascular dysfunction in patients undergoing invasive coronary angiography. Our search identified one randomised trial (CorMicA) evaluating stratified medical therapy guided by invasive physiological assessment, including coronary flow reserve, index of microcirculatory resistance, fractional flow reserve, and acetylcholine testing, in patients with angina and non-obstructive coronary artery disease, which showed improvement in angina severity at 6 months. We also identified a few observational studies evaluating the prognostic implications of coronary microvascular dysfunction. However, existing evidence is limited by retrospective designs, small sample sizes, selected populations, and heterogeneity in the definition of coronary microvascular dysfunction, which might limit its applicability to patients undergoing invasive coronary angiography in routine clinical practice.

Added value of this study

This prospective multicentre study provides evidence from a consecutive real-world cohort undergoing invasive coronary angiography with physiological assessment, showing that coronary microvascular dysfunction coexists with obstructive epicardial coronary artery disease and is associated with adverse clinical outcomes. These findings challenge the perception that coronary microvascular dysfunction is predominantly associated with patients without obstructive epicardial disease, and extend previous evidence suggesting that it might have prognostic relevance independent of epicardial coronary artery disease. By evaluating both epicardial coronary artery disease and coronary microvascular dysfunction in the same population, this study supports a more integrated pathophysiological framework for ischaemic heart disease and highlights the need for further investigation.

Implications of all the available evidence

Taken together with existing evidence, these findings suggest that coronary microvascular dysfunction might represent an important component of ischaemic heart disease that is not fully captured by assessment of epicardial coronary artery disease alone. Although the observational design precludes causal inference, these results support the potential value of incorporating microvascular functional assessment for risk stratification in patients undergoing invasive coronary evaluation. Further studies are needed to establish whether diagnostic and therapeutic strategies targeting coronary microvascular dysfunction can improve clinical outcomes and to define the role of integrated epicardial and microvascular assessment in routine clinical practice.

prognosis of coronary microvascular dysfunction in patients undergoing clinically indicated invasive coronary angiography in routine practice through a prospective, multicentre study.

Methods

Study design and patients

The Multicenter FLOW-CMD Registry was an investigator-initiated, prospective, multicentre cohort study done at seven tertiary hospitals in South Korea. We prospectively enrolled consecutive patients aged 18 years or older with suspected ischaemic heart disease who were referred for clinically indicated invasive coronary angiography. All patients with intermediate epicardial stenosis (40–90% by visual estimation) underwent comprehensive physiological assessment, including FFR, CFR, and IMR. Patients with multivessel disease and at least one lesion with intermediate stenosis (40–90%) were included. Patients were excluded if they met any of the following criteria: single-vessel disease

with lesions exceeding 90% diameter stenosis; clinically significant vasospasm (subtotal occlusion or >90% of diameter stenosis by ergonovine provocation test); cardiogenic shock or cardiac arrest; non-cardiac comorbid conditions and a life expectancy less than 2 years; inability to undergo coronary physiological assessment due to technical, administrative, or logistical reasons; pregnancy or breastfeeding; or being deemed ineligible for enrolment by treating physicians due to clinical instability, procedural complexity, or inability to undergo physiological assessment. Detailed inclusion and exclusion criteria are provided in the appendix (p 8). Eligible patients in whom all three physiological indices (FFR, CFR, and IMR) were obtained in at least one coronary artery were enrolled. Data on race and ethnicity were not collected. Sex was recorded as a biological variable (male or female) from medical records at enrolment.

The study protocol was approved by the institutional review board at each participating site (appendix p 6). All

patients provided written informed consent before enrolment. A Data and Safety Monitoring Board oversaw the study and an independent Clinical Event Adjudication Committee, whose members were masked to the study group assignments, assessed all clinical events (appendix pp 3–5). The study protocol and statistical analysis plan are available in the appendix (pp 48–201).

Procedures

Coronary physiological assessment was done after diagnostic angiography. Measurement protocols for FFR, CFR, and IMR were standardised and prespecified among the participating centres (appendix p 16). In brief, a pressure and temperature-sensor guide wire (Abbott Vascular; St Paul, MN, USA) was used. To derive resting mean transit time (Tmn), a thermodilution curve was obtained by delivering three injections (4 mL each) of room-temperature saline. After administration of hyperaemic agents, proximal aortic pressure (Pa), distal arterial pressure (Pd), and hyperaemic Tmn were measured. CFR is calculated as resting Tmn/hyperaemic Tmn. IMR is calculated as hyperaemic $Pd \times \text{hyperaemic Tmn}$ and is corrected in vessels with FFR of 0.80 or less (corrected IMR = $Pa \times Tmn \times [1.35 \times Pd/Pa] - 0.32$).¹² In patients with acute myocardial infarction, invasive physiological assessment was limited to non-culprit vessels and was performed after successful treatment of the infarct-related artery during the index procedure. For lesions with FFR of 0.80 or less or epicardial coronary artery disease with more than 90% stenosis, percutaneous coronary intervention (PCI) was recommended as per current guidelines^{1,2} with either biodegradable or biocompatible polymer-coated everolimus-eluting stents; however, a decision regarding PCI was at the discretion of operators (physicians). Intravascular imaging-guided stent optimisation was strongly recommended according to prespecified protocols (appendix pp 17–19). In patients treated by PCI, post-PCI physiological assessment was repeated. Obstructive epicardial coronary artery disease was defined as an intermediate stenosis (40–90% diameter stenosis) with FFR of 0.80 or less or severe stenosis (>90%) treated with revascularisation without FFR measurement. Coronary microvascular dysfunction was identified as CFR lower than 2.0 and IMR of ≥ 25 in any evaluated vessels at the index procedure. In patients who underwent PCI and in whom post-PCI physiological assessment was performed, CFR and IMR were measured after PCI was used to define coronary microvascular dysfunction. All angiograms and coronary physiological assessment data were analysed by the two independent core laboratories (appendix p 5).

Clinical follow-up was done during scheduled outpatient clinic visits at months 1, 6, and 12, and yearly thereafter. Patients unable to attend outpatient clinical visits were contacted by telephone. Cross-validation of

survival status was done with the Korean National Health Insurance database.

Outcomes

The primary endpoint was a patient-oriented composite outcome of all-cause death, myocardial infarction, clinically driven repeat revascularisation, or hospitalisation for heart failure. Secondary endpoints included individual components of the primary endpoint,

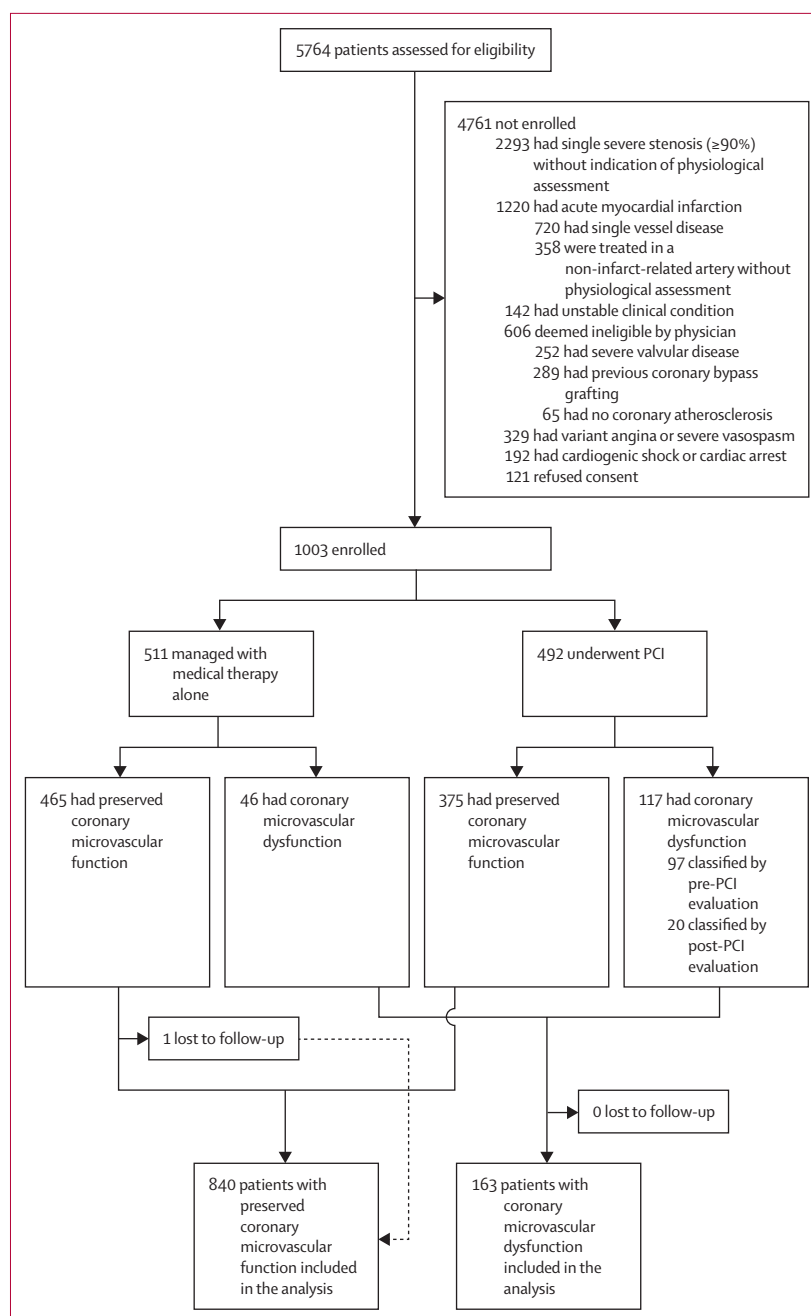


Figure 1: Study selection

PCI=percutaneous coronary intervention.

	Study population (n=1003)
Median age (years)	65·0 (58·0–72·0)
Sex	
Male	756 (75·4%)
Female	247 (24·6%)
Median BMI (kg/m ²)	24·5 (22·7–26·8)
Initial presentation	
Chronic coronary syndrome	664 (66·2%)
Acute coronary syndrome	339 (33·8%)
Unstable angina	200 (19·9%)
Acute myocardial infarction	139 (13·9%)
Non-ST-segment elevation myocardial infarction	86 (8·6%)
ST-segment elevation myocardial infarction	53 (5·3%)
Initial symptom	
Typical chest pain on exertion	635 (63·3%)
Atypical symptoms	368 (36·7%)
Medical history	
Hypertension	702 (70·0%)
Diabetes	406 (40·5%)
Insulin-treated diabetes	53 (5·3%)
Dyslipidaemia	870 (86·7%)
Current smoking	132 (13·2%)
Chronic renal insufficiency	104 (10·4%)
Previous percutaneous coronary intervention	194 (19·3%)
Previous myocardial infarction	86 (8·6%)
Previous stroke	104 (10·4%)
Peripheral arterial disease	22 (2·2%)
Atrial fibrillation or flutter	106 (10·6%)
Non-invasive tests before coronary angiography*	478 (47·7%)
Coronary CT angiography	422 (42·1%)
Exercise treadmill test	36 (3·6%)
Exercise stress echocardiography	21 (2·1%)
Single-photon emission CT	12 (1·2%)
Nitrogen-13 ammonia PET	8 (0·8%)
Cardiac MRI	4 (0·4%)
Non-invasive tests results	
Abnormal results	457/478 (95·6%)
Normal results	21/478 (4·4%)
Laboratory test	
Median LDL cholesterol (mg/dL)	74·0 (56·0–106·0)
Median glycated haemoglobin (%)	6·0 (5·7–6·7)
Median estimated creatinine clearance (mL/min per 1·73 m ²)	76·4 (57·9–97·2)
NT-proBNP (pg/mL)	125·0 (41·2–606·0)

(Table 1 continues in next column)

target vessel-related myocardial infarction, non-target vessel-related myocardial infarction, target vessel revascularisation, non-target vessel revascularisation, stroke, and Seattle Angina Questionnaire at baseline, 1 year, and 2 years after the index procedure. Vessels evaluated by coronary physiological assessment at the index procedure were defined as target vessels.

	Study population (n=1003)
(Continued from previous column)	
Median left ventricle ejection fraction (%)	62·7 (57·7–66·6)
Angiographic disease extent	
Insignificant disease	162 (16·2%)
Single-vessel disease	406 (40·5%)
Two-vessel disease	295 (29·4%)
Three-vessel disease	140 (14·0%)
Intervention	
Fractional flow reserve $\leq 0\cdot 80$ at least one vessel	444 (44·3%)
Percutaneous coronary intervention at least one vessel	492 (49·1%)
Drug-eluting stents	365/492 (74·2%)
Drug-coated balloon	69/492 (14·0%)
Both	58/492 (11·8%)
Intravascular imaging devices used	246/492 (50·0%)
Discharge medications	
Aspirin	565 (56·3%)
P2Y ₁₂ inhibitor	
Any	737 (73·5%)
Clopidogrel	637 (63·5%)
Ticagrelor	25 (2·5%)
Prasugrel	75 (7·5%)
Oral anticoagulant	71 (7·1%)
Statin	952 (94·9%)
Ezetimibe	741 (73·9%)
β blocker	358 (35·7%)
ACE inhibitor or ARB	574 (57·2%)
Calcium channel blocker	420 (41·9%)
Other angina medications (eg, trimetazidine, nicorandil)	302 (30·1%)
SGLT2 inhibitors	230 (22·9%)
Data are median (IQR) or n (%). ACE=angiotensin-converting enzyme. ARB=angiotensin receptor blocker. *25 patients had more than one non-invasive test.	

Table 1: Baseline characteristics

Definitions of secondary endpoints and clinical events are described in the appendix (pp 12–15).

Statistical analysis

We estimated that a sample size of 1003 patients would provide at least 80% power at a two-sided significance level of 5% to reject the null hypothesis of no difference in the primary endpoint between patients with preserved microvascular function and those with coronary microvascular dysfunction using the log-rank test given an anticipated accrual time of 2 years, follow-up duration of 1 year from last patient enrolment, and dropout rate of 0·5%. On the basis of previous studies (appendix pp 10–11),^{9,13,14} the annual incidence of the primary endpoint was expected to be 8·0% in patients with preserved microvascular function and 20·0% in patients with coronary microvascular dysfunction. The expected ratio

Lesions (n=1615)	
Location of lesion	
Left main artery	18 (1.1%)
Left anterior descending artery	977 (60.5%)
Left circumflex artery	288 (17.8%)
Right coronary artery	332 (20.6%)
Culprit lesion in acute coronary syndrome	323 (20.0%)
Pre-PCI TIMI flow grade (n=1248)*	
TIMI 2	9 (0.7%)
TIMI 3	1239 (99.3%)
Pre-PCI quantitative coronary angiography (n=1432)†	
Mean proximal reference diameter (mm)	3.4 (0.6)
Mean distal reference diameter (mm)	2.8 (0.6)
Mean minimal lumen diameter (mm)	1.6 (0.7)
Mean diameter stenosis (%)	49.7 (17.2)
Mean lesion length (mm)	25.9 (11.6)
Pre-PCI physiological characteristics (n=1248)*	
Mean resting Pd/Pa	0.93 (0.07)
Mean FFR	0.82 (0.10)
Mean resting Tmn (s)	0.93 (0.51)
Mean hyperaemic Tmn (s)	0.36 (0.21)
Mean CFR	2.9 (1.5)
Mean IMR	23.8 (13.1)
Procedural characteristics	
No PCI	873 (54.1%)
Suboptimal PCI	3 (0.2%)
Successful PCI	739 (45.9%)
Devices	
Drug-coated balloons	170/742 (22.9%)
Drug-eluting stents	567/742 (76.4%)
Both	5/742 (0.7%)
Mean device size (mm)	3.0 (0.5)
Mean number of stents	1.1 (0.4)
Mean length of stents (mm)	32.2 (13.9)
Post-adjunctive balloon	478/742 (64.4%)
Mean size of post-adjunctive balloon (mm)	3.3 (0.6)
Mean maximum pressure (atmosphere)	19.8 (4.6)

(Table 2 continues in next column)

Lesions (n=1615)	
(Continued from previous column)	
Intravascular imaging devices	
No intravascular imaging devices used	434/742 (58.5%)
IVUS	179/742 (24.1%)
OCT	129/742 (17.4%)
Timing of intravascular imaging devices	
Before PCI	16/308 (5.2%)
After PCI	34/308 (11.0%)
Both	258/308 (83.8%)
Post-PCI TIMI flow grade	
TIMI 3	742/742 (100%)
Post-PCI quantitative coronary angiography (n=689)‡	
Mean proximal reference diameter (mm)	3.3 (0.6)
Mean distal reference diameter (mm)	2.9 (0.5)
Mean minimal lumen diameter (mm)	2.5 (0.6)
Mean diameter stenosis (%)	14.9 (12.7)
Post-PCI physiological characteristics (n=347)§	
Mean resting Pd/Pa	0.93 (0.04)
Mean FFR	0.85 (0.06)
Mean resting Tmn (s)	0.79 (0.44)
Mean hyperaemic Tmn (s)	0.31 (0.17)
Mean CFR	2.8 (1.3)
Mean IMR	22.2 (13.2)

Data are n (%), mean (SD), or n/N (%). CFR=coronary flow reserve. FFR=fractional flow reserve. IMR=index of microcirculatory resistance. IVUS=intravascular ultrasound. OCT=optical coherence tomography. Pa=aortic pressure. Pd=distal coronary pressure. PCI=percutaneous coronary intervention. Tmn=mean transit time. TIMI=thrombolysis in myocardial infarction. *At baseline, coronary physiological assessment was not done in 367 lesions due to chronic total occlusion, critical stenosis with decreased coronary flow, culprit vessel of acute myocardial infarction, or technical reasons. Among 144 patients who underwent multivessel physiological assessment, 20 vessels had incomplete sets of CFR, IMR, or FFR measurements. These vessels were included in total lesion counts. The proportion of pre-PCI TIMI flow grades was presented only for the target vessels that were evaluated by invasive physiological assessment. †183 lesions and ‡53 lesions could not be analysed due to technical reasons such as vessel overlapping, insufficient contrast filling, or inaccurate calibration. §395 lesions were not evaluated by post-PCI coronary physiological assessment.

Table 2: Lesion and procedural characteristics

between preserved microvascular function and coronary microvascular dysfunction (sampling ratio) was 15:1 (ie, the incidence of coronary microvascular dysfunction was assumed to be approximately 6.7%), as suggested by a previous study.⁹ No interim analysis was planned.

All analyses were done on a per-patient basis for clinical characteristics and study endpoints and on a per-vessel basis for other factors. Categorical data were represented numerically and in terms of relative frequencies (percentages) and compared with the χ^2 test. Continuous variables were expressed as mean (SD) or median (IQR) and compared with the Student's *t* test or Mann-Whitney test according to distribution by Q-Q plot and Shapiro-Wilk test. Kaplan-Meier analyses were performed for time-to-event outcomes with between-group differences

estimated by Cox proportional hazard regression models, and results were presented as hazard ratios (HRs) with 95% CIs. The proportional hazards assumption was evaluated with a two-sided score test of the scaled Schoenfeld residuals over time at the 0.05 level. As a sensitivity analysis, we evaluated the cumulative incidence of the non-fatal clinical outcomes (myocardial infarction, repeat revascularisation, hospitalisation for heart failure, or stroke) using the Fine-Gray subdistribution hazards model, treating all-cause death as a competing event. For cardiac death, or a composite of cardiac death or myocardial infarction, non-cardiac death was treated as the competing risk.¹⁵ Prespecified subgroup analyses of the primary endpoint were done according to initial presentation, symptom status, treatment strategy for epicardial coronary artery disease, presence of functionally significant stenosis,

and clinical factors. All other subgroup and sensitivity analyses were considered exploratory and hypothesis-generating.

Baseline variables considered clinically relevant or showing a univariate association with the primary endpoint were entered into multivariate Cox proportional hazards regression models. To ensure model parsimony, variables were selected with consideration of the number of observed events,¹⁶ and non-significant variables were removed with backward selection. Continuous variables were modelled as linear terms in Cox models after formal assessment of the linearity assumption. Discrimination of multivariable Cox proportional hazards regression models was assessed by Harrell's c-index. No imputation methods were used for missing baseline variables. Statistical analyses were done with R version 4.5.1 and IBM SPSS Statistics for Windows version 25.0. The study protocol was registered with ClinicalTrials.gov (NCT05369182).

Role of the funding source

The study funders provided an investigator-initiated study grant for this research. However, they had no role in the study design, collection, analysis, and interpretation of data, or preparation of the manuscript.

Results

Between April 22, 2022, and Nov 19, 2024, 1003 patients with suspected ischaemic heart disease were enrolled after invasive coronary angiography with physiological assessment. Of 5764 patients screened, those with variant angina or severe vasospasm were excluded (figure 1). The median age of the study population was 65·0 years (IQR 58·0–72·0) and 756 (75·4%) were male and 247 (24·6%) were female (table 1). Overall, 664 patients (66·2%) presented with chronic coronary syndrome and 339 (33·8%) presented with an acute coronary syndrome. Typical chest pain on exertion was noted in 635 patients (63·3%). Before invasive coronary angiography, 478 patients (47·7%) had undergone non-invasive tests, with abnormal results in 457 (95·6%) of those tested. On invasive coronary angiography, 435 patients (43·4%) had multivessel disease, 406 (40·5%) had single vessel disease, and 162 (16·2%) had insignificant stenosis. By invasive physiological assessment, 444 patients (44·3%) had FFR of 0·80 or less in at least one vessel. PCI was performed in at least one vessel for 492 patients (49·1%), predominantly with drug-eluting stents. The use of antiplatelet, lipid-lowering, and other medications at discharge was high (appendix pp 27–28).

A total of 1615 lesions were evaluated. 1248 lesions were assessed with coronary physiological assessment and 347 lesions underwent post-PCI coronary physiological assessment (table 2). Target vessels had angiographically intermediate stenosis with diameter stenosis of 49·7% (SD 17·2) and FFR of 0·82 (SD 0·10). PCI was done in 742 (45·9%) of 1615 lesions, predominantly with drug-eluting stents. Intravascular imaging devices were used in 308 (41·5%) of 742 lesions and successful PCI was done in 739 (99·6%) of 742 lesions. No patients had serious adverse clinical events directly related to coronary physiological assessment, such as wire-induced dissection, distal embolisation, or intracoronary thrombus formation.

Among 1003 patients, 163 patients (16·3%; 271 vessels) had coronary microvascular dysfunction, and 840 patients (83·7%; 1344 vessels) had preserved microvascular function. Among 492 patients who underwent PCI, 117 patients had coronary microvascular dysfunction. Of these, 97 patients were identified as having coronary microvascular dysfunction based on pre-PCI evaluation, and only 20 patients showed coronary microvascular dysfunction in post-PCI measurements (appendix p 29). Among 144 patients with multivessel physiological assessment, 110 (76·4%) showed concordant results, whereas 34 (23·6%) showed discordant results regarding the classification of

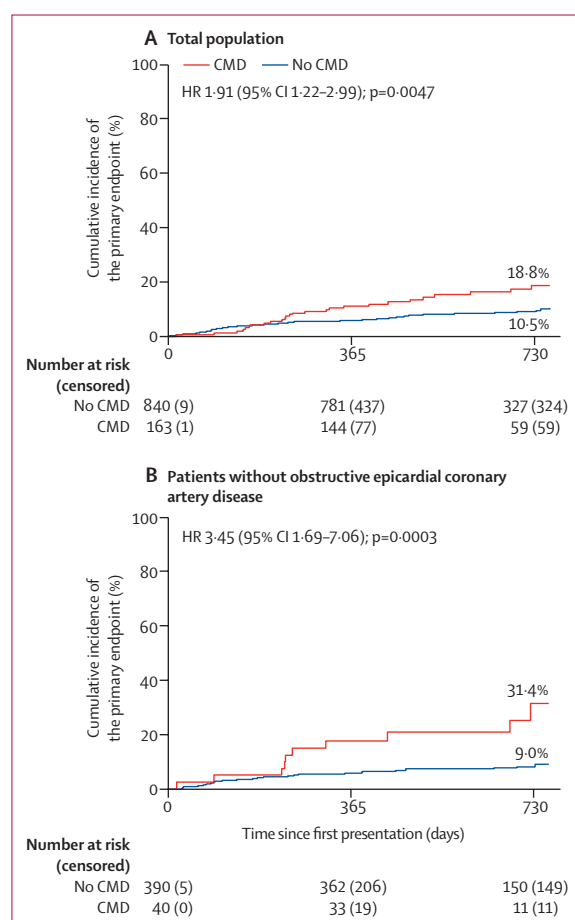


Figure 2: Cumulative incidences of primary endpoint
(A) Cumulative incidence for the total population. (B) The primary endpoint is compared according to the presence of coronary microvascular dysfunction among patients without epicardial coronary artery disease, defined by fractional flow reserve of 0·80 or more and deferred revascularisation. CMD=coronary microvascular dysfunction. HR=hazard ratio.

	Number of patients (2-year Kaplan–Meier estimates; %)			Absolute risk difference (95% CI)	Hazard ratio (95% CI)	p value
	Total (n=1003)	No CMD (n=840)	CMD (n=163)			
Primary endpoint						
Patient-oriented composite outcome*	96 (11.8%)	70 (10.5%)	26 (18.8%)	8.3 (1.1 to 15.6)	1.91 (1.22–2.99)	0.0047
Secondary endpoints†						
All-cause death	33 (4.8%)	23 (4.4%)	10 (7.7%)	3.5 (–1.6 to 8.6)	2.20 (1.05–4.61)	0.038
Cardiac death	20 (2.9%)	12 (2.2%)	8 (6.5%)	4.3 (–0.3 to 9.0)	3.38 (1.38–8.26)	0.0077
Non-cardiac death	13 (1.9%)	11 (2.0%)	2 (1.2%)	–0.8 (–3.1 to 1.4)	0.91 (0.20–4.12)	0.91
Cardiac death or myocardial infarction	31 (4.0%)	22 (3.5%)	9 (6.9%)	3.5 (–1.3 to 8.2)	2.08 (0.96–4.51)	0.065
Myocardial infarction‡	12 (1.3%)	10 (1.3%)	2 (1.2%)	–0.1 (–1.9 to 1.9)	1.03 (0.23–4.69)	0.97
Target vessel-related myocardial infarction	9 (1.0%)	8 (1.0%)	1 (0.6%)	–0.4 (–1.8 to 1.0)	0.64 (0.08–5.13)	0.68
Non-target vessel-related myocardial infarction	3 (0.3%)	2 (0.2%)	1 (0.6%)	0.4 (–0.9 to 1.7)	2.59 (0.23–28.53)	0.44
Deferred vessel myocardial infarction	8 (0.9%)	7 (0.9%)	1 (0.6%)	–0.3 (–1.7 to 1.1)	0.74 (0.09–5.98)	0.77
Repeat revascularisation§	39 (4.8%)	27 (4.0%)	12 (8.8%)	4.9 (–0.3 to 10.0)	2.30 (1.17–4.55)	0.016
Target vessel revascularisation	33 (4.0%)	25 (3.6%)	8 (6.1%)	2.4 (–2.0 to 6.9)	1.66 (0.75–3.68)	0.21
Non-target vessel revascularisation	12 (1.6%)	6 (1.0%)	6 (4.6%)	3.6 (–0.2 to 7.4)	5.21 (1.68–16.16)	0.0042
Deferred vessel revascularisation	27 (3.3%)	18 (2.7%)	9 (6.5%)	3.8 (–0.6 to 8.2)	2.60 (1.17–5.78)	0.019
Hospitalisation for heart failure¶	28 (2.9%)	21 (2.5%)	7 (4.9%)	2.4 (–1.5 to 6.3)	1.71 (0.73–4.03)	0.22
Stroke (ischaemic or haemorrhagic)	20 (2.1%)	19 (2.4%)	1 (0.6%)	–1.8 (–3.5 to –0.2)	0.27 (0.04–2.02)	0.20

Data are n (%) unless otherwise stated. The database for the analysis was locked on Nov 18, 2025. Clinical endpoints were evaluated during the overall study period. Percentages are the Kaplan–Meier estimates at 2 years. CMD=coronary microvascular dysfunction. PCI=percutaneous coronary intervention. *The number needed to harm regarding primary endpoint was 12.0 (95% CI 1.0–15.6). †The individual endpoints listed are the first occurrence of that event. ‡Myocardial infarction was defined with the Fourth Universal Definition of Myocardial Infarction.²⁷ Deferred vessel myocardial infarction is the myocardial infarction occurring in the vessel with deferred PCI at index procedure. §Repeat revascularisation includes all first clinically driven elective, urgent, or emergent revascularisation procedures that were not planned during the index hospitalisation during the overall study period. Target vessel denotes the vessel that was evaluated by physiological assessment at index procedure. Deferred vessel revascularisation is the revascularisation occurring at the vessel with deferred PCI at index procedure. ¶Hospitalisation for heart failure should include all of these criteria: (1) hospitalisation with primary diagnosis of heart failure, (2) duration of hospitalisation of at least 12 h, (3) new or worsening symptoms of heart failure, (4) objective evidence of new or worsening heart failure on physical examination or laboratory findings, and (5) initiation or intensification of heart failure treatment.

Table 3: Cumulative rates of primary and secondary endpoints

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coronary microvascular dysfunction. Patients with coronary microvascular dysfunction were older, more frequently had typical chest pain on exertion, and had higher rates of hypertension and chronic renal insufficiency than did those with preserved microvascular function (appendix pp 30–31). The rate of abnormal results on non-invasive tests did not differ between the two groups. N-terminal pro B-type natriuretic peptide concentration was higher in patients with coronary microvascular dysfunction than in those with preserved microvascular function. Patients with coronary microvascular dysfunction more frequently had FFR of 0.80 or less (60.7% vs 41.1%) and underwent PCI in at least one vessel (71.8% vs 44.6%) than did those with preserved microvascular function. The lesion-level analysis also showed similar results (appendix pp 32–34).

When stratified by the presence of obstructive epicardial stenosis, coronary microvascular dysfunction was observed in 123 (21.5%) of 573 patients with obstructive epicardial coronary artery disease (FFR ≤0.80 or PCI was performed for severe stenosis without FFR measurement) and in 40 (9.3%) of 430 patients without obstructive epicardial coronary artery disease (FFR >0.80 and deferred revascularisation).

At a median follow-up of 1.9 years (IQR 1.1–2.1), the primary endpoint was more frequent in patients with coronary microvascular dysfunction than in those with preserved microvascular function (HR 1.91 [95% CI 1.22–2.99]; $p=0.0047$; figure 2A; table 3), occurring in 26 patients (2-year Kaplan–Meier estimate 18.8%) with coronary microvascular dysfunction and 70 patients (2-year Kaplan–Meier estimate 10.5%) with preserved microvascular function. Among 430 patients without obstructive epicardial coronary disease, the primary endpoint was more frequent in patients with coronary microvascular dysfunction (HR 3.45 [95% CI 1.69–7.06]; $p=0.0003$; figure 2B), occurring in ten patients (2-year Kaplan–Meier estimate 31.4%) with coronary microvascular dysfunction and 30 patients (2-year Kaplan–Meier estimate 9.0%) with preserved microvascular function.

Individual components of the primary and secondary endpoints are shown in table 3. Cardiac death occurred in eight patients (2-year Kaplan–Meier estimate 6.5%) with coronary microvascular dysfunction and 12 patients (2-year Kaplan–Meier estimate 2.2%) with preserved microvascular function. The causes of death are summarised in the appendix (p 35). Patients with

coronary microvascular dysfunction had higher rates of clinically driven repeat revascularisation and deferred vessel revascularisation events than those with preserved microvascular function. The results of the competing risk analysis were consistent with the primary analysis results (appendix pp 36–37). Similar results were observed when coronary microvascular dysfunction was defined with pre-PCI assessment only (appendix pp 20, 38–39), or when an alternative cutoff value for depressed CFR (<2.5) was applied (appendix p 21).

Although the total Seattle Angina Questionnaire score did not differ between groups, patients with coronary microvascular dysfunction showed worse angina-related outcomes, with lower angina stability and lower angina frequency scores than those with preserved microvascular function at 2-year follow-up in patients with chronic coronary syndrome (appendix pp 22, 40–41).

On multivariable analysis, coronary microvascular dysfunction was independently associated with the primary endpoint in the overall population (HR 1.65 [95% CI 1.02–2.66]; $p=0.041$) and patients without obstructive epicardial coronary artery disease (3.64 [1.69–7.85]; $p=0.0003$; appendix p 42).

Prespecified and exploratory subgroup analyses showed consistent results across all subgroups including initial presentation (chronic coronary syndrome or acute coronary syndrome), symptom status (typical chest pain on exertion or atypical symptoms), age, sex, presence of diabetes, chronic renal insufficiency, previous non-invasive testing (none, coronary CT angiography, or stress testing), and the presence of functionally significant epicardial stenosis (appendix pp 23–24, 43). Although a significant interaction was observed between treatment strategy for epicardial coronary artery disease (medical therapy alone vs PCI) and the presence of coronary microvascular dysfunction regarding the risk of primary endpoint ($p_{\text{interaction}}=0.034$), these findings should be interpreted with caution given the small number of events and the exploratory nature of the analyses (appendix p 23).

Regarding different endotypes of coronary microvascular dysfunction, 356 (35.5%) of 1003 patients showed discordant classification between CFR and IMR (221 patients had $\text{CFR} \geq 2.0$ and $\text{IMR} \geq 25$ and 135 patients had $\text{CFR} < 2.0$ and $\text{IMR} < 25$; appendix p 44). However, patients with CFR less than 2.0 and IMR of ≥ 25 (known as structural coronary microvascular dysfunction) showed a consistently higher risk of primary endpoint than the other groups (appendix p 25).

Discussion

The Multicenter FLOW-CMD Registry study shows that coronary microvascular dysfunction often coexists with epicardial coronary artery disease in patients undergoing clinically indicated invasive coronary angiography with physiological assessment for suspected ischaemic heart disease. At a median follow-up of 1.9 years, coronary microvascular dysfunction was associated with a higher

risk of the composite of all-cause death, myocardial infarction, clinically driven repeat revascularisation, or hospitalisation for heart failure compared with preserved microvascular function.

Coronary microvascular dysfunction reflects functional or structural abnormalities of the coronary microcirculation and can cause myocardial ischaemia.⁴ Although coronary microvascular dysfunction can occur in patients with various cardiovascular conditions, it is most frequently encountered in patients with ischaemic heart disease.⁴ Current guidelines suggest coronary physiological assessment with CFR and IMR in patients with ischaemic heart disease but without clinically significant epicardial coronary artery stenosis.^{5,6} These recommendations are based on previous reports indicating that coronary microvascular dysfunction is associated with worse clinical outcomes.^{7–11,18} However, existing evidence is limited by retrospective designs,⁷ small sample sizes,^{8,9} selected populations,^{9–11} and heterogeneous definitions of coronary microvascular dysfunction,^{19,20} and prospective data in unselected patients undergoing invasive coronary angiography remain scarce.

We prospectively screened more than 5000 consecutive patients with suspected ischaemic heart disease who were referred for invasive coronary angiography in routine practice and applied prespecified protocols of coronary physiological assessment to evaluate the presence of coronary microvascular dysfunction. Among the 1003 patients who underwent the assessments, 16.3% had coronary microvascular dysfunction. In our study, the prevalence of coronary microvascular dysfunction was 9.3% among patients without obstructive epicardial coronary artery disease and 21.5% among those with obstructive epicardial coronary artery disease. Although current guidelines suggest evaluating coronary microvascular dysfunction primarily in patients without obstructive coronary artery disease, our findings suggest that it also coexists with epicardial coronary stenosis and could also be relevant in patients with more complex or functionally significant disease. Patients with coronary microvascular dysfunction had numerically worse angina profiles after PCI, although differences were not statistically significant. This observation is consistent with previous studies suggesting coronary microvascular dysfunction as a potential mechanism underlying persistent angina after PCI.^{21,22} Therefore, evaluation of coronary microvascular function might help to identify mechanisms underlying patients' symptoms beyond epicardial coronary artery disease, particularly in the context of clinical presentation and suspected ischaemic heart disease.

In our study, patients with coronary microvascular dysfunction had a higher risk of the composite primary endpoint than those with preserved microvascular function. Several mechanisms might explain this association. In addition to myocardial ischaemia, coronary microvascular dysfunction is linked to endothelial dysfunction and inflammation, which can

contribute to atherosclerotic progression and coronary vasomotor abnormalities.¹⁹ Another potential mechanism is diffuse myocardial fibrosis, as chronic microvascular impairment can lead to repetitive ischaemia, interstitial collagen deposition, and adverse ventricular remodelling.²³

The association between coronary microvascular dysfunction and the risk of the primary endpoint appeared to vary by treatment strategy for epicardial disease, with a more pronounced association in patients receiving medical therapy alone than in those undergoing PCI. However, these interaction analyses were exploratory and should be interpreted with caution given the small number of events. These findings suggest that coronary microvascular dysfunction might identify a phenotype of potential prognostic relevance, particularly in the absence of flow-limiting epicardial disease. In patients treated with PCI, the attenuated association might reflect competing mechanisms or procedure-related microvascular alterations (appendix p 26). This finding suggests that a treatment paradigm focused on epicardial coronary stenosis might not fully capture the prognostic relevance of coronary microvascular dysfunction. Assessment of coronary microcirculation during invasive coronary angiography might provide complementary information for risk stratification and identify patients with residual risk despite revascularisation and optimal medical therapy. Together, these results support an integrated framework of coronary artery disease incorporating both epicardial and microvascular pathology. Further studies are needed to establish whether targeting coronary microvascular dysfunction improves clinical outcomes.

The limitations of our study should also be considered. First, coronary physiological assessment results were not masked to treating physicians, which might have influenced subsequent management. However, potential bias was mitigated by prespecified endpoint definitions, blinded core laboratory assessment, and independent adjudication of clinical events. Interpretation should nevertheless be cautious, particularly in patients without epicardial coronary artery disease, given the limited statistical power in this subgroup. Second, the study population comprised patients with intermediate epicardial coronary artery stenosis undergoing invasive physiological evaluation for suspected ischaemic heart disease; therefore, individuals with normal or minimal disease were not systematically included. The generalisability of these findings to patients with normal coronary arteries remains uncertain and warrants further study. Third, coronary physiological assessment was mostly done in a single vessel rather than across all major coronary territories, and coronary microvascular dysfunction might exhibit spatial heterogeneity. Any resulting misclassification is likely to be non-differential and might bias the associations towards the null. Therefore, the prognostic signal observed in our cohort might be underestimated rather than overstated. Fourth, the study population was recruited exclusively from centres in

South Korea, and regional practice patterns might influence patient selection and therefore the observed prevalence of epicardial coronary artery disease and coronary microvascular dysfunction, limiting generalisability (appendix p 45). Fifth, coronary physiological assessment adds cost to angiography or PCI, and its cost-effectiveness requires further evaluation. Sixth, in some patients, coronary microvascular dysfunction was defined based on the post-PCI physiology assessment. However, defining coronary microvascular dysfunction based on post-PCI physiological assessment might have introduced misclassification. Nevertheless, sensitivity analyses based on pre-PCI assessment showed consistent results. Seventh, vasomotor testing with acetylcholine was not done. Patients with vasospastic angina were assessed with ergonovine provocation and excluded if clinically significant vasospasm was present. Therefore, patients with combined vasospasm and coronary microvascular dysfunction were not represented. Finally, physiological assessments were done without a standardised washout of vasoactive or anti-anginal therapies, reflecting real-world practice. Accordingly, results represent on-treatment physiology, and medication effects might have attenuated observed associations.

In conclusion, coronary microvascular dysfunction defined by CFR less than 2.0 and IMR of 25 or more coexisted with epicardial coronary artery disease in patients with suspected ischaemic heart disease undergoing invasive coronary angiography, and patients with coronary microvascular dysfunction had a higher risk of the composite endpoint of all-cause death, myocardial infarction, clinically driven repeat revascularisation, or hospitalisation for heart failure. Although causal inference is limited by the observational design, these findings suggest that coronary microvascular dysfunction might represent an important component of ischaemic heart disease beyond epicardial coronary artery disease. Further studies are warranted to establish whether targeted diagnostic and therapeutic strategies for coronary microvascular dysfunction can improve clinical outcomes.

Contributors

All authors participated in the study design, enrolment of patients, and contributed to clinical follow-up. JML, SHL, DHG, SYL, and KHC analysed and interpreted the data. JML and SHL designed the study, wrote the first and subsequent drafts of the manuscript, and made the decision to submit the manuscript for publication. JML, SHL, and SYL accessed and verified all the underlying data. All authors approved the final version of the manuscript. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Declaration of interests

JML reports funding from Abbott Vascular, Boston Scientific, Microport, and Terumo Corporation; and consulting fees from Abbott Vascular, Boston Scientific, and Pulse Medical. SHL reports funding from Abbott Vascular. KHC reports funding from the Korean Society of Cardiology and Abbott Vascular. YBS reports funding from the Korean Society of Cardiology and Microport. J-YH reports funding from the National Evidence-based Healthcare Collaborating Agency, the Ministry of Health & Welfare (Korea), Abbott Vascular, Biosensors, Biotronik, Boston Scientific, Daiichi Sankyo, Dong A-ST, Hanmi Pharmaceutical, and Medtronic. All other authors declare no competing interests.

Data sharing

Participant consent for data sharing was not obtained and therefore individual participant data collected for this study will not be made publicly available. However, de-identified individual participant data will be available upon reasonable request for the purpose of collaborative research. Proposals should be directed to the corresponding author (JML; drone80@hanmail.net) and will be reviewed on the basis of scientific merit and feasibility. Data will be available from the publication date for a period of 5 years. Access will be provided after approval of a proposal and with a signed data access agreement. The study protocol and statistical analysis plan of the Multicenter FLOW-CMD Registry are available in the appendix (appendix pp 48–201).

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